

M003 Week 7 — English Worksheet

Topic: Designing the Museum · **Course:** 3D Coordinates · **Time:** about 45 minutes

A museum brings together everything in this course: a **floor** laid out with (x, z) , **pictures on the walls** drawn with (x, y) , and **sculptures on pedestals** placed with (x, y, z) . This week you make a **blueprint** — you decide the coordinates of every piece **before** you build, so nothing overlaps and every sculpture sits on its stand.

1 · Predict 🧠

Read each plan. Before you imagine it happening, write what you think you will see.

```
floor:    fill stone from (0, 0, 0) to (8, 0, 8)
picture:  (drawn on the back wall, where  $z = 0$ , using  $x$  and  $y$ )
pedestal: fill stone from (2, 0, 2) to (4, 1, 4)
```

The floor is flat (y stays 0). The picture stands up the back wall. The pedestal is a short box. Looking down from above, do the picture and the pedestal sit in different places? Why won't they crash?


```
pedestal A: fill stone from (1, 0, 1) to (2, 1, 2)
pedestal B: fill stone from (6, 0, 6) to (7, 1, 7)
```

Two pedestals. Read their (x, z) footprints. Are they far apart or on top of each other? How do you know from the numbers?


```
pedestal: fill stone from (2, 0, 2) to (4, 1, 4)
statue:   place gold block at (3, 2, 3)
```

The pedestal fills up to height 1. The statue is placed at height 2. Does the statue rest on the pedestal, float above it, or sink into it?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The statue should rest **on top** of the pedestal. The pedestal fills up to height 1, so the statue belongs at height 2. Right now it floats high in the air.

```
fill stone from (2, 0, 2) to (4, 1, 4)
place gold block at (3, 5, 3)
```

Hint: the top of the pedestal is one block above where the stone ends.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — These two pedestals should stand in **different corners** of the floor. Right now they share the same (x, z) footprint and pile up in one spot.

```
fill stone from (1, 0, 1) to (2, 1, 2)
fill stone from (1, 0, 1) to (2, 1, 2)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This picture should **stand up the back wall**, where z stays 0 and the blocks climb with x and y. Right now it is lying flat on the floor.

```
set x to 0
repeat 4 times:
  place red block at (x, 0, 0)
  add 1 to x
set x to 0
repeat 4 times:
  place red block at (x, 0, 1)
  add 1 to x
```

Hint: a wall picture grows **up** (y), it does not step **forward** (z). The second row should be one block higher, not one block forward.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Draw a **blueprint** of one museum room **before** you build. Looking down from above, mark the floor, the back wall picture, and at least **two pedestals** — and write the coordinates of each.

Floor: from (... , ... , ...) to (... , ... , ...)

Pedestal 1: from (... , ... , ...) to (... , ... , ...) · its statue at (... , ... , ...)

Pedestal 2: from (... , ... , ...) to (... , ... , ...) · its statue at (... , ... , ...)

Now build your room from the blueprint, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I planned ...

I kept my two pedestals from overlapping by ...

Each statue sits on its pedestal because its height starts at ...

My wall picture stands up because it grows along ... not ...

One tricky moment was when ...

If I had more time, I would ...

First, I planned ...

I kept my two pedestals from overlapping by ...

Each statue sits on its pedestal because its height starts at ...

My wall picture stands up because it grows along ... not ...

One tricky moment was when ...

If I had more time, I would ...

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One tricky moment was when ...

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4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk through your museum room. Talk through it like you are giving a tour. Try to use these words: **blueprint, floor, pedestal, sculpture, overlap.**

1. Show your blueprint on paper, then show the built room.
2. Point at two pedestals and say their (x, z) footprints, and why they don't overlap.
3. Show a statue resting on its pedestal and say the height it starts at.
4. Say in your own words how planning coordinates first made the room work.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.